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Fire-safety aspects of long-span timber structures in industrial buildings

S.L. Benner, X. Browne & O. Popovic Larsen
Royal Danish Academy: Architecture, Design, Conservation, Institute of Architecture and Technology, Copenhagen, Denmark

ABSTRACT: This paper investigates the possibilities and challenges in relation to fire-safety for using timber as a structural material in industrial buildings. Combining a literature study with interviews with fire-safety specialists, it goes through historic and fire-safety aspects of industrial buildings, as well as the intrinsic fire properties of virgin and reused timber. The findings show that the prerequisites to a high extent have changed for this building typology, making it often more suitable and safer to apply timber structures than prior to the 20th century where industrial buildings had significant fire issues. The choice between virgin and reused timber is not without importance though: conventional timber components with large cross sections have some beneficial and predictable intrinsic properties in relation to fire where reused timber components, with more complex compositions, can be less predictable and would therefore depend highly on different fire-protection systems, such as sprinkling and/or coverings.

1 INTRODUCTION

This paper investigates the opportunities and challenges for using timber (virgin and reclaimed) in contemporary factory and warehouse buildings (FW-buildings) in relation to fire. Timber is renowned historically as a material for structural applications in factory buildings. Prior to the 20th century, timber was one of the primary structural materials used in industrial buildings, but, due to the risk of fire, timber was later substituted with non-combustible materials such as concrete, iron, and steel. From the second half of the 19th century steel and concrete received increased interest from the building industry and therefore timber was largely abandoned as a construction material alongside its related knowledge from builders and designers (Crocetti 2016).

The Intergovernmental Panel on Climate Change calls for a significant reduction of humans' impact on the earth's ecological systems, where a substantial part of the global environmental impact is related to the energy and building sector (IPCC, 2023). Today, FW-buildings are being built to produce components used for the transition to renewable energy, such as windmills, solar panels, etc. While these buildings are most often built from concrete or steel, The United Nations Environmental Panel points to solutions including the use of reused and bio-based materials for climate mitigation (UNEP 2022). Here, timber as a construction material is experiencing a revitalization with structural solutions with engineered wood such as cross laminated timber (CLT), glue laminated timber (GLT), laminated veneer lumber (LVL). However, with six of the nine planetary boundaries exceeded, including biosphere integrity and land-system change, holistic solutions are needed for reducing the human impact on the ecological systems including minimizing the deforestation tendencies globally (Richardson et al. 2023). Therefore, the objective of the study is to investigate the opportunities and challenges in relation to fire-safety for using virgin and/or reused timber for load-bearing structures in today's FW-buildings and to present design considerations for both practice and research. The research question investigated in this paper is:

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- What are the opportunities/challenges for using timber (both virgin and reused) in long-span structures in contemporary factory and warehouse buildings in relation to fire?

2 METHODS

The paper is divided into two parts: (1) a review of the use of timber-structures and aspects of fire-safety in historic industrial buildings, and (2) an examination of fire-safety aspects in FW-buildings and material properties of timber structures. To develop a more comprehensive understanding, the study adopts a qualitative research strategy, combining a literature review with semi-structured interviews. The literature review was conducted using JSTOR and Google Scholar, employing various search terms such as "timber structures," "fire safety," "timber connections," and "industrial buildings". The literature selection focused on aspects of passive fire properties of timber structures and its connections, as well as the impact from active fire-protection systems.

For the interviews, four fire-safety specialists (three engineers and one architect working with fire-properties of timber) were selected based on their extensive knowledge about fire-safety in timber structures and/or FW-buildings, ensuring a cross-disciplinary perspective. This approach aligns with research suggesting that interdisciplinary collaboration enhances architectural studies and fosters alternative modes of thinking (Sattrup 2011). The interviews were conducted using a semi-structured format, allowing for new insights and previously unconsidered aspects. To guide the discussions, two fictional timber structures (Figure 1) is introduced and were presented to the interviewees, representing a FW-building with a 40-meter span, 15-meter height, and 10-meter center-to-center spacing. Various fire-protection solutions (Figure 2) were also presented to facilitate expert analysis. The study focuses on two structural timber systems: trapezoidal Warren trusses, suitable for 30–80 meter spans and typically made of GLT or LVL (Crocetti 2016), and the Hilding Brosenius system with a higher complexity in relation to fire-safety because of its composition. The latter has previously been used in a 47-meter-span sports hall designed by Alvar Aalto in 1952 (Pajunen, Crocetti, and Browne 2024). Following Groat and Wang's (2013), the interview data was systematically analyzed, organized, and synthesized with the literature review to develop a meaningful and well-supported understanding of fire safety in FW-buildings.

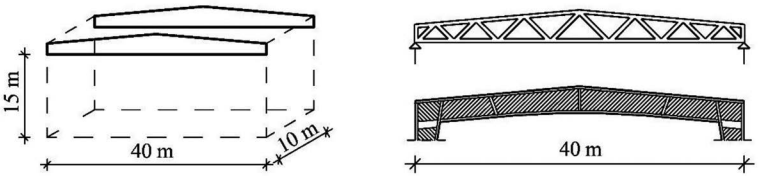


Figure 1. Left: the fictional room dimensions. Right: Warren truss (top) and Brosenius system (bottom).

3 REVIEW OF HISTORIC FACTORY BUILDINGS

As pointed out by Jevremovic, Vasic, and Jordanovic (2012), efficiency and safety have been essential aspects for the development of industrial buildings where fire-safety, cost, and flexibility greatly influenced the designs of these buildings. Therefore, this section focuses on the development of factory buildings and the impact from fire-safety and adaptability aspects.

3.1 *Early mills: A fire hazard*

The literature implies that factory buildings originate back to mills of the early 18th century with John Lombe's five-story high silk mill in Derby, from 1718, as an early example (Pierson,

1949; Winter 1970). It was made with a simple open-plan construction which, in factory and mill buildings, changed little during the next two hundred years (Winter 1970, 24). In general, the mills were designed in relation to the machines needed for production and were typically made as compact, multi-story buildings for keeping costs down and efficiency up. The buildings were repetitive and rhythmic where materials appeared raw and visible. They consisted of regular framed timber structures on the inside and with load-bearing masonry walls as facades. With a long and narrow form, they had open floor plans and a great number of windows for ensuring efficient daylight and workspace (Jevremovic, Vasic, and Jordanovic 2012).

Old mills, such as water and steam mills for cotton and silk production, had significant fire issues. Many mills burned down and were seen as dangerous fire hazards with fire spreading vertically as well as horizontally due to the large area of unprotected timber from the frames, floors, and rafters. Winter (1970) notes that casualties were surprisingly few and that it was surprising that the mills didn't burn down on a weekly basis since the cotton mills were dealing with poor ventilation, cotton everywhere, overhead shaft covered in oil, candle lights, and with young children and overworked people working in the building to name a few. But the large sized timber components used for the framing made it hard for the fire to take hold, which, Winter further notes, perhaps was the only reason for the buildings not to burn down more frequently (Winter 1970).

3.2 *Fire-proof mills*

An increasing number of fires in the mill buildings made the mill builders/engineers look for new fire-safety solutions. In some cases, timber beams were covered with sheet metal and plaster. In others, starting in the end of the 18th century, cast-iron columns and beams became a non-combustible alternative to the timber with brick vaulting in between the beams, replacing the timber ceilings. However, due to cast-iron being brittle and subject to sudden cracking in a case of fire, especially when water is thrown at it, fatal incidents started to occur. For this reason, many mill builders went back to the use of timber framing throughout the 19th century. Here, the most efficient timber mill was made following the 'slow burning principle' (Winter 1970, 47–52).

Wermiel (2004) points out that 'slow burning principle' - held by an association of fire-insurance companies in America - supplied novel solutions for heavy timber construction in factory buildings. Starting in the late 19th century, the aim for the standard was to provide solutions with maximum fire-safety and minimum costs. Because it was too expensive to build the mills using non-combustible materials, such as concrete and steel, they developed a novel system using the traditional materials: masonry and timber. Solutions ranged from using large-dimension timber members and non-combustible materials to automatic sprinkling. The flooring consisted of thick timber planks spanning between the beams with layers of water-proofing paper on top of the planks, and then hardwood flooring as finish which provided a fire- and waterproof floor. The compact timber frames and flooring proved to be efficient, due to how the timber would char instead of bursting out into flames. For preventing vertical and horizontal spread of fire, shafts and floor openings were closed, and large open floors were subdivided with firewalls. Eventually the attics were abandoned due to their inaccessibility and unprotected wooden rafters, where the high-pitched roofs were changed to roofs with limited slope (Wermiel 2004).

3.3 *Factories from the 20th century to today*

Entering the early 20th century the technologies and production methods developed further, and buildings had to adapt for new production needs. More adaptable solutions were obtained by using long-span structures of concrete and steel in single-story buildings. Here, Peter Behrens single-story AEG turbine factory from 1909 is often mentioned as an important classic example of early modern architecture and was constructed with large steel frames and glass walls, providing the factory with a significant open space and daylight (Winter 1970). Albert Kahn was also to play a significant part in the development of industrial architecture,

where he is described as being revolutionary within industrial architecture (Hildebrand 1975). With the aim to keep cost and construction time down, he refined the repetitive steel framed structure which also enabled better conditions for natural ventilation and daylight. Single-story solutions for factory and warehouse buildings are still in a Danish context the most common today, but as one of the interviewees notes, it's not rare to still use multi-story solutions. Also, in today's factories and warehouses there is a great variation of use, where some still depend on the presence of humans and others are driven by robots. Some FW-buildings still require natural daylight, but often there is little or no need at all. With the understanding of the historic development of factory and warehouse buildings, it's important to understand today's requirements, prerequisites, design considerations, etc. for being able to assess the use of timber structures in today's solutions. Therefore, the next section presents aspects of the material properties of timber in relation to fire and the requirements/prerequisites for FW-buildings through interviews and a literature review.

4 ASSESSMENT OF INTERVIEW AND LITERATURE DATA

4.1 *Requirements*

Load-bearing structures in a European context are designed to fulfill a required classification, such as REI60/A2-s1,d0. Starting with the REI60 part: the 'R' stands for *load-bearing capacity*, the 'E' for *integrity*, the 'I' for *isolation*, and the number indicates how many minutes the different categories must uphold their technical abilities in a fire scenario. The second part, A2-s1,d0, indicates the material fire properties. The materials reaction to fire are divided into the following primary classes: A1, A2, B, C, D, E, or F where A1 is the highest requirement. Two additional classes are made for smoke (s) and flaming drops (d). The class for smoke is either s1 (very limited amount of created smoke), s2 (limited amount of created smoke), and s3 (no requirement for created smoke). The flaming drops are categorized as either d0 (none), d1 (limited), and d3 (without requirement for burning drops of particles). D-s2,d0 is often used for GLT.

In our scenario the load-bearing structure consists of either trusses or frames and the *integrity* and *isolation* parameters are therefore not included. In addition to the Danish building regulations, there are pre-accepted solutions for fire-safety strategies for different building typologies. The listed requirements for the load-bearing structures in industrial and warehouse buildings are either 'no requirement'; R30; R60; R15/A2-s1,d0; R30/A2-s1,d0; or R60/A2-s1, d0. The classification depends on factors such as fire load, the fire growth factor, potential stacking height of inventory, fire-section area, and the potential implementation of active fire prevention systems, such as sprinkling and fire ventilation. Smaller facilities often have good opportunities for implementing timber structures, two interviewees add, where they often have been used in sports halls. Also, if e.g. a factory simply needs a large room volume for production needs, as in windmill or steel production, there might also be good opportunities. It would though be unwise to implement combustible construction materials in facilities with a lot of combustible inventory stacked in heights.

4.2 *Material properties*

Timber has some beneficial intrinsic fire properties compared to steel, the latter of which can be difficult to predict in a fire scenario. When timber is exposed to temperatures around 100 degrees Celsius the water starts to evaporate, whereafter, at around 300 degrees Celsius, a formation of charr occurs. The charr layer can't be calculated with any structural properties, but it isolates and protects the inner-timber from the heat which can thereby uphold its structural properties (Domański and Kmiecik 2019). The charring rate of timber is higher in the initial phases of a fire, where approx. 25 mm of charring is needed to be effective for isolating (Friquin 2011; Östman et al. 2010). The charring rate is otherwise relatively linear with European softwoods listed as 0.65 mm/min. If a timber component is exposed on three or four sides, such as beams and columns, there will be an increased impact due to the occurrence of rounded burn-in

formations (Figure 2). A notional charr rate of 0.7 mm/min for GLT and 0.8 mm/min for solid timber can be used for components (softwoods) exposed to a fire on three or four sides. Behind the charr layer there is a zone affected by the heat with reduced strength and stiffness. Though approx. 35-40 mm of the cross section on each side is effected, it is for reasons of simplification that only 7 mm is included as the so called 'zero-strength layer' (Östman et al. 2010).

These intrinsic properties of timber are highly beneficial for components with a larger cross section, as often the case with conventional glulam elements. A sufficient protection layer can be obtained by simply increasing the cross-section of the members by multiplying the charr rate with the required time and adding the 'zero-strength layer'. In relation to this, an interesting aspect is brought up by two of the interviewees: because a certain amount of structural bending can be allowed in a fire scenario in some cases, a reduced fire-protecting layer can be tolerated. This is caused by the 'over dimensioning' of the members for avoiding bending in none-fire situations. By calculating a reduced stiffness of the members in a fire scenario, the members would here bend when the cross-section is reduced by the fire, but without collapsing within the required time for evacuation. In some scenarios bending cannot be allowed though, e.g. if the member is placed above a window which would then fracture or if the bending will damage the sprinkling system.

The Brosenius system is more challenged than conventional GLT or LVL members due to its small cross section, area of exposed surfaces, exposed ends, unevenness, and cracks. This could potentially be mitigated by increasing the cross section with at least 25 mm on each exposed side, as with the conventional timber components, and by covering the exposed ends of the members (Figure 2). But, as two of the interviewees add, a significant amount of precision and processing would be needed for avoiding cracks and unevenness, which would be highly time consuming. Instead, one interviewee suggests that Brosenius beams could be spanning between the load-bearing structure as either purlins or rafters in roof elements covered by e.g. gypsum. This could also enable the use of reclaimed timber as members in the Brosenius system, where unevenness to a greater extent can be tolerated. An alternative solution could be to cover each component with either gypsum or a 'sacrificial' timber layer. This 'sacrificial' layer could potentially be made of reclaimed timber instead of virgin timber and thereby create a technical layer with new aesthetic possibilities, with varying expressions from the different reclaimed timber sources.

Char rates are interesting to explore in relation to the use of reclaimed timber as a fire protecting 'sacrificial layer'. When sourcing different reclaimed timber products there might be a variation in timber species and properties. A generalization is that hardwoods have a slower charr rate with a more compact charr layer than softwoods, such as spruce and pine, which ignites easier, burns faster, and releases more heat due to its porous, cellular structure (Zang et al. 2024). The advantageous fire properties of high-density timber species have been known for thousands of years where Vitruvius in *De Architectura* wrote that larch, with its high density and less porous structure, could be useful as a construction material for preventing the spread of fire in buildings (Vitruvius Pollio; translation, Richard Schofield 2009). This is substantiated by Friquin (2011) who found that there is a broad acceptance by researchers that the charring rate for timber decreases with increasing density. Eurocode 5 presents a linear rate of 0.65 mm/min for softwoods, beech of any density, and hardwoods under 290 kg/m^3 and 0.50 mm/min for hardwoods with a density of more than 450 kg/m^3 (Liu and Fischer 2024). One interviewee mentions that their current research shows that the charr rate of reused timber is similar to virgin timber, but that it can be challenging to determine the species of the different sourced timber. Also, the quality of the timber is important to have in mind, where cracks and knobs will have a negative effect on the fire properties. Therefore, they are suggesting a more general charr rate for reclaimed timber.

Another crucial aspect to consider is the area of exposed surfaces, emphasized by three of the interviewees as well as Zang et al. (2024). This is caused by the ignition of the combustible surfaces releasing both energy and toxic smoke to the room in a fire situation. In Denmark, an often-used strategy is to limit the amount of exposed timber surfaces in a room to 20% (excluding floors). This can be a reasonable strategy when working with timber, but, as two of the interviewees mentions, the placement of the timber surfaces and the function of the room

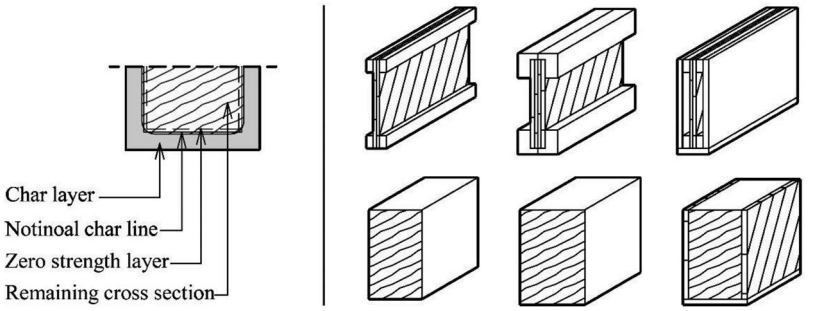


Figure 2. Left: the different layers of a timber cross-sections in a fire. Right: The Brosenius beam and conventional GLT member with no added fire-protection (left), increased cross-section (and covered ends for the Brosenius beam) (middle), and covering of either gypsum, plywood, waste wood, etc. (right).

have a more significant impact on a fire. This is also accounted for in the Danish regulations, one interviewee adds, where the 20% exposed surfaces must be 'evenly distributed' in warehouses and factory buildings. Two contradicting examples could be a hotel room with exposed CLT walls facing each other with little distance - where the ignition and radiation of the walls will uphold and intensify the fire - and a large-scale timber frame structure with adequate spacing between the frames where the fire wouldn't spread directly from one frame to another.

4.3 *Fire detection and extinguishing*

Automatic sprinkling, detection, and ventilation can be effective ways for preventing or mitigating fires. Zang et al. (2024) found that crucial time for evacuation and fire countermeasures is provided by technological advancement and data analysis for detection systems. Frangi and Fontana (2005) discovered that automatic sprinkling greatly limited the development of a fire in a hotel room even in a room with combustible surfaces, interior, and coverings. The fire spread rapidly where automatic sprinkling was not used and the increased room temperature stood as a challenge, which correlates with the interviewee's remarks. This is further substantiated by Zheng, Garis, and Pike (2022) which found that smoke alarms and sprinklers play a more important role in reducing fires than the choice of material, even in multi-story buildings.

In Danish FW-buildings sprinkling must be used if the fire-section is more than 2.000 m^2, one interviewee adds. If the section is e.g. only 1.800 m^2, it's also possible to use fire ventilation instead of sprinkling. This is sometimes preferred for economic reasons, but from a fire engineering perspective one adds that sprinkling is to be preferred over fire ventilation. In these two scenarios, with either sprinkling or fire ventilation, the R15/A2-s1,d0 requirement can be used due to the active fire-protection systems. The R15 requirement, another interviewee adds, is a relatively new 'specialty' for FW-buildings in the Danish building regulations and can only be used for buildings with either sprinklers or fire ventilation. Normally in Denmark active fire-prevention systems won't provide a 'discount' for the fire-performance of the load-bearing structure. This is because the load-bearing structures are designed to withstand a fire even if the sprinkling is not activated. In other countries there are significantly lower requirements for the load-bearing structures if the buildings have sprinklers, one interviewee adds. The A2-s1,d0 part of the R15 requirement excludes the use of timber, even in buildings with sprinklers. Timber probably could be included in the R15 requirement, one of the interviewees responded when asked, but that it probably hasn't been considered. Two of the interviewees agreed that there might be limited challenges for using timber structures in modern mega-factories. One interviewee stated that in large-scale halls it's close to being to being without matter if the frames are of timber or steel if the placement and amount of the timber components are considered.

5 DISCUSSION

Three aspects have been found essential to discuss in relation to this study: (I) change in building composition and layout from the old mills until today's FW-buildings, (II) intrinsic material properties of timber and the impact from active fire-prevention systems, and (III) the use of reused timber. (I) Today the risk of fire spread seems significantly lower than in the old FW-buildings prior to the 20th century which had considerable fire issues. Back then the buildings were typically compact, multi-story buildings, with many exposed timber surfaces. These are ideal conditions for vertical spread of fire, which can be considered a major factor for moving away from timber as a construction material in FW-buildings. In contradiction, many of today's solutions depend on modern fire detection and extinguishing systems, large room volumes, large spans, and single-story solutions without attics. In that sense, the use of load-bearing timber structures can seem more reasonable from a fire-safety perspective due to the adequate spacing and relatively low amount of exposed timber surfaces. However, this still depends on the function of the building, the inventory, and the placement of the timber components.

(II) The use of timber components with larger cross sections are favorable due to timber's intrinsic fire properties and the simplicity of calculation methods. Therefore, it's relatively simple to use the trapezoidal Warren truss, where the members are often made of conventional GLT. It must though be considered that all four sides are exposed in a truss system, increasing the impact from the fire. The Brosenius system seems more challenged in general, where the slender cross-section stands as a considerable challenge. This could be mitigated by increasing the cross-section of the beam, but the beam's complexity – with cracks, knobs, and exposed ends - makes it difficult to predict in a fire and thereby also to calculate its fire properties. It is therefore necessary to carry out different fire tests for the Brosenius system for determining its fire properties. The complex composition of the Brosenius system could be solved with coverings, but this will increase both cost and construction time. Also, visible structures have been a main characteristic in historic industrial buildings articulating the simple rhythmic and repetitive systems. In a Danish context, sprinkling can only give a 'discount' for non-combustible structures in industrial buildings, but it seems unclear whether this requirement can be applied for timber in the future as well. Often in these large factories the need of sprinkling will already be present due to the large room volumes, where the sprinklers therefore wouldn't add as an additional cost. However, this opens for several other questions: what if the active systems don't work? What if the function, and thereby the requirements, of the room changes? How will regional regulatory differences influence the possible use of timber? Here it might be necessary to re-assess whether some fire-safety requirements should be updated for determining the potential use of different timber structures in FW-buildings.

(III) The use of reused timber in relation to fire is challenged by the uneven surfaces, which would need a significant amount of processing and precision for making fire-proof members. Here it can be rational for both environmental and technical reasons to place them where they are covered and protected, such as if used as purlins or rafters in roof elements. This will require less timber and enables 'imperfections' of reclaimed timber to remain. There might be scenarios where it would be desirable to have the reused timber members exposed for aesthetic reasons, but even though the fire properties of reused timber are considered equal to virgin timber, this would probably still highly depend on the use of active fire extinguishing systems because of the high complexity in the composition of the reused timber members. An alternative could be to use reused timber as a 'sacrificial' covering of the load-bearing structure with either virgin or reused timber instead, but this solution would probably face the same time-consuming issue for obtaining the required precision. A central last question therefore remains also: how will a Brosenius beam or coverings made of reclaimed timber perform in a fire? For this, fire-tests need to be carried out for different reused timber components for determining their structural properties in a fire.

6 CONCLUSION

This study suggests that there often are enhanced conditions for using conventional timber structures in today's FW-buildings. The large room volumes, large spans, fire-protection

systems, etc. makes it considerably less challenging compared to FW-buildings prior to the 20th century. Timber with larger cross-sections has some favorable intrinsic properties in relation to fire, but the placement of the components, the area of exposed timber surfaces, and the function of the room are also crucial aspects to consider. Furthermore, it seems more challenging to have exposed timber components with more slender cross-sections and high geometric complexities, as with the Brosenius system. These challenges can potentially be mitigated by providing a 'discount' for the structures if the building is supplied with an automatic sprinkling system. This is not typically allowed in a Danish context, but is, though, allowed for non-combustible structures in industrial buildings. Reused timber components are challenged by the complexity in how they are assembled, where fire-tests must be carried out to determine the fire properties of them. As an alternative, they can potentially be used as secondary components protected by coverings, such as rafters in roof elements or purlins, minimizing complexity, risk, and timber consumption.

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